Beichen Xue

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RESEARCH INTEREST

My research lies at the intersection of eye tracking, visualization, and computational modeling of attention and interactive behavior. I study how gaze and multimodal interaction traces reflect task-driven user behavior and develop cognitively grounded models for predicting user scanpaths and evaluating interfaces.

EDUCATION

- National University of Singapore, 2021-2025
  - Bachelor of Science in Data Science, Minor in Computer Science

PUBLICATIONS

- **Beichen Xue**, Yue Jiang. Toward Human-Like Task-Driven Gaze Navigation for Visualization. Under review at CHI 2027.
- Xiangyu Zhou, **Beichen Xue**, Yue Jiang, Steve Oney. Pace by Gaze: A Gaze-Informed Simulator of Constrained Cursor Movement. Under review at CHI 2027.
- Yura Hwang, **Beichen Xue**, Yin Yang, Yue Jiang. DashTrace: Understanding Task-Driven Visual Attention and Multimodal Behavior in Analytical Dashboards. Under review at CHI 2027.
- Jinghui Hu, **Beichen Xue**, Hans Gellersen, Yue Jiang. Gaze in the Era of Foundation Models: A Survey of Integration, Interaction, and Evaluation. Under review at CHI 2027.
- Katja Pott, Eunhye Kim, Chenyang Wang, Thiemo Wambsganss, Juho Kim, …, **Beichen Xue**, … Matthias Söllner (50+ authors). Generative User Interfaces: A Definition and Design Space. Under review at CHI 2027.
- Parvin Emami, **Beichen Xue**, Yue Jiang, Luis A. Leiva. How Human Is Human-Like? Interpreting Scanpath Similarity Metrics. Under review at ETRA 2027.
- Kewen Peng*, **Beichen Xue***, Meng Chen, Felix Kretzer, Jeffrey Nichols, Tiffany Knearem, Alexander Maedche, Christof Lutteroth, Hans Gellersen, Jeffrey P. Bigham, Yue Jiang. Attention Is All UI Needs: The 6th Computational UI Workshop at CHI. Workshop proposal under review at CHI 2027. (*Equal contribution)
- Jingshu Li, Tianqi Song, **Beichen Xue**, Yi-Chieh Lee. We Shape AI, and Thereafter AI Shape Us: Humans Align with AI through Social Influences, ICLR 2025 Workshop & CHI 2025 SIG.
- Shida Wang, **Beichen Xue**. State-Space Models With Layer-Wise Nonlinearity Are Universal Approximators With Exponential Decaying Memory, NeurIPS 2023.

SKILLS

- Machine Learning/Deep Learning: Reinforcement Learning, VIsion-Language Models, Bayesian Inference, PyTorch-Transformers, Graph Learning
- System Development: Full-Stack (Frontend + ML-based backend) interactive system development
- Physiological Signals: Eye-tracking and EEG Signal
- Database: PostgreSQL (PL/pgSQL)

EXPERIENCES

Technical HCI Research Assistant, **Dec 2025 -**
University of Utah, School of computing. **Advisor: Prof. Yue Jiang** **Present**

- Designed and conducted task-driven eye movement data collection studies to capture how users attend to and interpret visualization.
- Developed models to simulate human gaze behavior, incorporating task context and cognitive activities.

HCI Research Intern,
National University of Singapore, School of Computing. **Advisor: Prof. Yi-Chieh Lee** **Nov 2024 – Aug 2025**

- Designed experiments on exploring how AI confidence levels expressed through natural language impact human confidence in decision making context. Investigated verbal and cognitive alignment phenomena between human and AI.
- Designed EEG-based experiment to assess cognitive states of human confidence before and after decision-making.
- Explored social influence of AI through contagion and conformity in interactions with humans.

Machine Learning Research Project,
National University of Singapore, Department of Mathematics. **April 2023– June 2023**

- Assisted in mathematically establishing incorporating layer-wise nonlinear activation into stacked State-Space Models improves their ability to approximate complex sequence-to-sequence relationships.
- Demonstrated the exponential memory decay of state-space models through empirical analysis.

Machine Learning Intern, Amaris.AI **May 2022 – July 2022**

- Conducted evaluation of a T5 transformer model for question generation tasks
- Investigated application of Knowledge Graph Ontology to enhance generation of abstract and templated questions, improving model's ability to create varied and contextually relevant outputs.